

PCCP

Accepted Manuscript



This article can be cited before page numbers have been issued, to do this please use: C. A. Cardenas, F. Heidar-Zadeh and P. Ayers, *Phys. Chem. Chem. Phys.*, 2016, DOI: 10.1039/C6CP04533B.



This is an *Accepted Manuscript*, which has been through the Royal Society of Chemistry peer review process and has been accepted for publication.

Accepted Manuscripts are published online shortly after acceptance, before technical editing, formatting and proof reading. Using this free service, authors can make their results available to the community, in citable form, before we publish the edited article. We will replace this *Accepted Manuscript* with the edited and formatted *Advance Article* as soon as it is available.

You can find more information about *Accepted Manuscripts* in the [Information for Authors](#).

Please note that technical editing may introduce minor changes to the text and/or graphics, which may alter content. The journal's standard [Terms & Conditions](#) and the [Ethical guidelines](#) still apply. In no event shall the Royal Society of Chemistry be held responsible for any errors or omissions in this *Accepted Manuscript* or any consequences arising from the use of any information it contains.

Benchmark Values of Chemical Potential and Chemical Hardness for Atoms and Atomic Ions (Including Unstable Anions) from the Energies of Isoelectronic Series

Carlos Cárdenas^{a,c,d,*}, Farnaz Heidar-Zadeh^{b,e,f} and, Paul W. Ayers^b

- a) Departamento de Física, Facultad de Ciencias, Universidad de Chile, Casilla 653, Santiago 7800024, Chile .
- b) Department of Chemistry & Chemical Biology, McMaster University, Hamilton, Ontario L8S 4M1, Canada.
- c) Centro para el Desarrollo de la Nanociencia y la Nanotecnología (CEDENNA), Avda. Ecuador 3493, Santiago 9170124, Chile.
- d) Temporary address: Universidad Nacional de Colombia - Sede Medellín, Facultad de Ciencias, Escuela de Física.
- e) Department of Inorganic & Physical Chemistry, Ghent University, Krijgslaan 281 (S3), 9000 Gent, Belgium
- f) Center for Molecular Modeling, Ghent University, Technologiepark 903, 9052 Zwijnaarde, Belgium

Abstract.

We present benchmark values for the electronic chemical potential and chemical hardness from reference data for ionization potentials and electron affinities. In cases where the energies needed to compute these quantities are not available, we estimate the ionization potential of the metastable (di)anions by extrapolation along the isoelectronic series, taking care to ensure that the extrapolated data satisfies reasonable intuitive rules to the maximum possible extent. We also propose suitable values for the chemical potential and chemical hardness of zero-electron species. Because the values we report are faithful to the trends in accurate data on atomic energies, we believe our proposed values for the chemical potential and chemical hardness are ideally suited to conceptual studies of chemical trends across the periodic table. Critical nuclear charge (Z critical) of the isoelectronic series with $2 < N < 96$ are also reported for the first time.

I. Introduction

The simplest possible model for electron transfer between chemical species requires two parameters, one that measures the tendency of a substance to accept electrons and one that measures the substance's resistance to donating electrons. These two parameters can be identified with the electron affinity and ionization potential or, somewhat more intuitively, with an electronic chemical potential that measures the energetic benefit/detriment of gaining/losing electrons¹ and a chemical hardness that measures how strongly the system resists being in its optimal charge stage.² The electronic chemical potential is inversely related to the electronegativity, since (a) systems with high electronegativity have large (in magnitude) negative chemical potentials and (b) Sanderson's electronegativity equalization principle is a direct consequence of the equalization of electronic chemical potential in a stable molecule.^{1,3,4} The chemical hardness is also related to fundamental conceptual rules, most notably the hard/soft acid/base principle and the maximum hardness principle.⁵⁻²⁵

The chemical potential and chemical hardness are venerable concepts, dating back to at least the 1960's. However, it was only around 1980 that a rigorous theoretical framework, based on density functional theory (DFT), for the electronic chemical potential and chemical hardness was constructed. In this framework, the change in a system's electronic energy, E , with respect to the number of electrons, N , at fixed molecular geometry (encapsulated in the external potential binding the electrons, $v(\mathbf{r})$) is expanded as a Taylor series,^{2,26}

$$E[v(\mathbf{r}); N_0 + \Delta N] = E[v(\mathbf{r}); N_0] + \left(\frac{\partial E[v(\mathbf{r}); N_0]}{\partial N} \right)_{v(\mathbf{r})} \Delta N + \frac{1}{2} \left(\frac{\partial^2 E[v(\mathbf{r}); N_0]}{\partial N^2} \right)_{v(\mathbf{r})} (\Delta N)^2 + \dots \quad (1)$$

The first two coefficients in the Taylor series are identified as the electronic chemical potential,¹

$$\mu = \left(\frac{\partial E}{\partial N} \right)_{v(r)}, \quad (2)$$

and the chemical hardness²

$$\eta = \left(\frac{\partial \mu}{\partial N} \right)_{v(r)} = \left(\frac{\partial^2 E}{\partial N^2} \right)_{v(r)} . \quad (3)$$

Higher-order terms are normally neglected as insignificant, though there has been a recent resurgence of interest in them.²⁷⁻³¹

The mathematical framework provided by Eq. (1) allows one to quantify the chemical potential and chemical hardness. One merely requires a *model* for the dependence of the energy on the number of electrons. (The exact result for an isolated substance is a piecewise linear curve, which is clearly nondifferentiable.^{28, 32, 33} So the goal is to find a model that explains the dependence of the energy on the number of electrons *in a chemical environment*.³⁴⁻³⁸) Numerous models have been proposed in the literature,^{26, 39-42} but the simplest model, and the only one that is directly compatible with the common decision to ignore third- and higher-order derivatives in the Taylor series expansion, is a quadratic model, $E(N) = a + bN + cN^2$. It is elementary to show that the chemical potential and hardness can then be expressed directly in terms of the vertical ionization potential, IP, and vertical electron affinity, EA,^{1, 2}

$$\mu(N_0) = -\frac{1}{2}(\text{IP}(N_0) + \text{EA}(N_0)) \quad (4)$$

$$\eta(N_0) = \text{IP}(N_0) - \text{EA}(N_0) . \quad (5)$$

where

$$\text{IP}(N_0) = E[v(\mathbf{r}); N_0 - 1] - E[v(\mathbf{r}); N_0] \quad (6)$$

$$\text{EA}(N_0) = E[v(\mathbf{r}); N_0] - E[v(\mathbf{r}); N_0 + 1] \quad (7)$$

Another reason to choose the quadratic model is that this model lets the chemical potential and chemical hardness of an infinite system be identified with the Fermi level and the fundamental gap, respectively. We note that the first- and higher-order derivatives of E , μ , and η with respect to the external potential provide useful information about which locations within a molecule are most reactive and/or sensitive to *alchemical changes*.⁴³ In this regard, it is worth to note that the quadratic energy model results in properly normalized (non-)local descriptors in contrast to other common energy models.^{44,45-55}

The major goal of this paper is to provide a comprehensive tabulation of the electronic chemical potential and chemical hardness for atoms and atomic ions. This

extends and modernizes existing tables of this type,^{2, 56, 57} and to a large extent merely requires the collection, curation, and combination of a vast quantity of benchmark data. However, there are some atomic ions (and, rarely, neutral atoms) that require a more thoughtful treatment. For example, the proton, H^+ , has zero electrons and no ionization potential. However, given its ubiquity in chemistry, it seems necessary to provide a sensible value for the chemical potential and hardness of the proton. Similarly, an anion like F^- has no known electron affinity; the oxygen dianion, O^{2-} , has neither a known electron affinity nor a known ionization potential.

Any strategy we use for defining the chemical potential and chemical hardness of such species is obviously somewhat arbitrary, but we believe that the advantage of providing *reasonable* values for the chemical potential and hardness of these species is nonetheless valuable. Moreover, since these values are mainly used for *qualitative* studies of relative chemical reactivity, the primary objective of our approach is to provide values that faithfully reproduce chemical trends: the precise numerical values of these reactivity indicators should not be overinterpreted as, indeed, even when the values of the chemical potential and hardness seem unambiguous, they still depend on our decision to *choose* the quadratic model for $E(N)$.

In section II of this paper, our approach to compute the chemical potential and the chemical hardness is explained. Sections III and IV present our results and summarize various (un)successful trials. All the data is provided as supplementary material.

II. Data and Methods

A. Sources of Data

An extensive tabulation of the ionization potential of neutral atoms and positively-charged atomic ions is curated by the U.S. National Institute of Standards and Technology (NIST). We confirmed that these values closely resemble those of the 88th Edition of the CRC Handbook of Chemistry and Physics, and the choice of values does not affect our results. We also cross-referenced the NIST ionization potential data against the NIST atomic spectra database, ensuring that there were no disagreements.^{58, 59} In a few cases, we detected discrepancies between the NIST data and the original references,

and the values from the original references were used in these cases, all of which corresponds to such highly charged cations as to be chemically irrelevant.

The data for electron affinities was taken from the 88th Edition of the CRC Handbook of Chemistry and Physics. Once again, the difference between the electron affinity values reported by different sources is too small to affect our results significantly. We therefore feel confident that the reported values can be considered reliable benchmark values, and are not likely to change significantly.

B. Method for Zero-Electron Atomic Ions

Zero-electron ions have no ionization potential, so the Parr-Pearson formula cannot be invoked. To find values for the chemical potential and chemical hardness of the proton, we used the asymptotic formulas,^{13, 60}

$$\mu(Z) = aZ^{-1/3} \quad (8)$$

$$\eta(Z) = bZ^{-1/3} \quad (9)$$

and regressed against the values for the alkali metal cations, Li⁺ to Cs⁺. This gives the values

$$\begin{aligned} \mu[\text{H}^+] &= -56 \pm 4 \text{ eV} & (R^2 = .995) \\ \eta[\text{H}^+] &= 92 \pm 12 \text{ eV} & (R^2 = .985) \end{aligned} \quad (10)$$

A similar treatment for the Helium dication (the α -particle) gives

$$\begin{aligned} \mu[\text{He}^{2+}] &= -90 \pm 25 \text{ eV} & (R^2 = .97) \\ \eta[\text{He}^{2+}] &= 135 \pm 50 \text{ eV} & (R^2 = .93) \end{aligned} \quad (11)$$

For the Helium dication, we used the alkaline earth dications, Be²⁺ to Sr²⁺, to perform the regression, since this gave better correlations than the noble gas dications. We do not know instances where the Lithium trication or other, more highly-charged, zero-electron species arise in chemistry, so we did not extend our treatment to those systems.

The values in Eqs. (10)-(11) seem reasonable: the proton and the α -particle are both strong acids and extremely hard (they are extremely small, and they resist giving up electrons to an infinite extent). One could argue that the chemical potential values are too negative, but they are consistent with the value of chemical potential of He^+ , -40 eV.

C. Method for Atoms/Ions that Do Not Bind an Electron

When an atom or ion does not bind an additional electron, its electron affinity is formally zero, because (cf. Eq. (7)) $E_{g.s.}[\nu(\mathbf{r}); N_0 + 1] = E_{g.s.}[\nu(\mathbf{r}); N_0]$. This assumption is not obviously unsatisfactory, but it was already recognized in the initiating paper of Parr and Pearson on the chemical hardness that using the energy of an energy resonance for the anion, thereby defining a negative electron affinity, also gave sensible results.^{2, 61-64} The motivation for that choice is that this metastable anion state would be stabilized in the presence of a solvent, and that the properties of this state would therefore be relevant when, as is typically the case, gas-phase values for the chemical potential and hardness are used to describe the behavior of atoms and functional groups of atoms in solvent, or within a molecule.⁵⁷ The problem with this approach—aside from practical difficulties associated with obtaining an accurate value for the energy of the metastable state—is that there can be several metastable states, and it is not certain whether the lowest-energy or the longest-lived resonance is the chemically relevant state.

Our strategy is to instead choose the value for the (negative) electron affinity to extend the *trends* that are present in the chemical potential, hardness, ionization potential, and other quantities for stable atomic systems. This seems especially appropriate for descriptors, like the chemical potential and chemical hardness, that are primarily used for qualitative elucidation of chemical trends.

We start by noting that the energy of atoms and ions can be written as a Laurent series in terms of the atomic number/nuclear charge, Z , and the number of electrons, N ,^{13, 43, 65-71}

$$E(N, Z) = F_0(N)Z^2 + F_1(N)Z + F_2(N) + F_3(N)Z^{-1} + \dots \quad (12)$$

where the coefficients of the expansion, $F_k(N)$, only depend on N , and thus are the same for every element in an isoelectronic series. Because Z is a very large number, we find it more convenient to express this series in terms of the atomic charge, $q = Z - N$,

$$E(N, q) = f_0(N)q^2 + f_1(N)q + f_2(N) + f_3(N)(q + N)^{-1} + \dots \quad (13)$$

The ionization potential (cf. Eq. (6)) satisfies a similar series,

$$\text{IP}^{(N)}(q) = p_0^{(N)}q^2 + p_1^{(N)}q + p_2^{(N)} + p_3^{(N)}(q + N)^{-1} + \dots \quad (14)$$

where superscript (N) is used to emphasize that the number of electrons is a parameter, rather than a variable, for an isoelectronic series. The electron affinity, chemical potential, and chemical hardness have similar expansions,

$$\text{EA}^{(N)}(q) = \text{IP}^{(N+1)}(q-1) = a_0^{(N)}q^2 + a_1^{(N)}q + a_2^{(N)} + a_3^{(N)}(q + N)^{-1} + \dots \quad (15)$$

$$\mu^{(N)}(q) = -\frac{1}{2} \left(\text{IP}^{(N+1)}(q-1) + \text{IP}^{(N)}(q) \right) = m_0^{(N)}q^2 + m_1^{(N)}q + m_2^{(N)} + m_3^{(N)}(q + N)^{-1} + \dots \quad (16)$$

$$\eta^{(N)}(q) = \text{IP}^{(N)}(q) - \text{IP}^{(N+1)}(q-1) = h_0^{(N)}q^2 + h_1^{(N)}q + h_2^{(N)} + h_3^{(N)}(q + N)^{-1} + \dots \quad (17)$$

Although the expansion coefficients for μ and η are directly related to the expansion coefficients for the ionization potential, when the coefficients are determined by fitting to experimental data, the expected relationships between the coefficients will not hold.

The key ingredient in our approach is to fit these expansion coefficients to accurate data for the isoelectronic series, and then extrapolate to obtain the corresponding quantities for atomic ions that are not physically realizable. Consider, for example, the five-electron isoelectronic series. The Beryllium anion does not exist, but its electron affinity (and, consequently, the chemical potential and hardness of the neutral beryllium atom) might be obtained by extrapolation, using known data for $q = 0, 1, 2, 3, 4$, etc., corresponding to B, C^+ , N^{+2} , O^{+3} , F^{+4} , etc.. This strategy may not provide incredibly accurate electron affinities (though in cases where the electron affinity exists, the results are quite satisfactory), but it provides chemically sensible values, consistent with the trends associated with isoelectronic series.

There are several different strategies that can then be envisioned.

- (i) Fit/extrapolate IP and EA for the isoelectronic series, and then use the definitions in equations (4) and (5) to determine μ and η .

- (ii) Construct μ and η for systems with positive electron affinities, and then directly fit/extrapolate μ and η .
- (iii) Combine (i) and (ii) approaches, for example, one can fit/extrapolate IP and μ , then determine the hardness with

$$\eta = 2(\text{IP} + \mu) \quad (18)$$

To determine the best strategy for fitting/extrapolation, one needs to assess the reliability of the predicted values for μ and η . Since we do not have reference values for atoms and ions that do not bind additional electrons, we instead assess our models based on their ability to reproduce certain features that reasonable chemical potential and hardness scales are believed to possess. For example:

- (a) Because the energy is a convex function of the number of electrons, the hardness should be positive.^{28, 32, 72-79} (There is no mathematical proof of convexity, but there are no experimental or theoretical counterexample for stable atoms and atomic ions.) We believe that $\eta < 0$ should occur rarely, if at all. The only case we can envision where $\eta < 0$ might be acceptable is where a (metastable) anion corresponds to a highest-occupied molecular orbital (HOMO) that is partially filled and degenerate, while the $N+1$ electron resonance corresponds to a favorable half-filled or wholly-filled HOMO level.
- (b) The hardness decreases as the charge decreases. That is, $\eta(+1) > \eta(0) > \eta(-1) > \eta(-2)$. Based on the known cases where this trend is in question, there may be some exceptions associated with half-filled and filled-shell cases (atoms and ions with zero total angular momentum).²⁷
- (c) The chemical potential increases as the charge decreases, $\mu(-2) > \mu(-1) > \mu(0) > \mu(+1)$. Positive values are allowed, and expected for species with unbound electrons because metastable states where one or more electrons occupy resonance states that have negative ionization potentials and also (as inferred from (a)) negative electron affinities.

D. Fitting Procedure and Objective Function

In order to carry out the fitting procedure, one needs to decide how many terms in the series, Eqs. (14)-(17), to consider. It is known that the series converges rapidly, and that a quadratic model (truncation after three terms) suffices to fit the ionization potential of the atoms in the first five rows of the periodic table.⁸⁰ So, we consider the quadratic model and the model with one additional term, $(q + N)^{-1}$. In our tests, we saw no benefit to including additional terms.

To develop a satisfactory fitting procedure, we first test the ability of the fitted ionization potentials to predict the (known) electron affinities of neutral atoms. (Neutral atoms that do not bind an additional electron were omitted from this analysis.) Linear least-squares fitting of the expansion coefficients in Eq. (14) to seven ionization potentials (for $q = -1, 0, 1, 2, \dots, 6$) gave good fits ($R^2 > 0.99$), but both the quadratic regression model and the $(q + N)^{-1}$ model failed to give satisfactory predictions for the electron affinities of the neutral atoms. (See Figure 1.) This occurs because the fit is strongly biased towards the ionization potentials of the highly-charged cations (with IPs of hundreds of eV), so employing a weighted least-squares fit is more reasonable. After some trial and error, we select an objective function which minimizes the relative error, with extra emphasis on species with small charge,

$$\min_{\{p_k^{(N)}\}} \sum_q \left(\frac{\text{IP}_{\text{ref}}^{(N)}(q) - \text{IP}^{(N)}(q, p_k^{(N)})}{\text{IP}_{\text{ref}}^{(N)}(q)(q + N)} \right)^2 \quad (19)$$

where $\text{IP}_{\text{ref}}^{(N)}(q)$ is the accurate ionization potential from the literature and $\text{IP}^{(N)}(q, p_k^{(N)})$ is the quadratic or higher-order expression from Eq. (14), with its dependence on the linear coefficients $p_0^{(N)}$, $p_1^{(N)}$, $p_2^{(N)}$, and (sometimes) $p_3^{(N)}$ made explicit. This gives excellent results for neutral atoms; see Figure 2. The values for the electron affinity from extrapolation, $\text{EA}_{\text{fit}}^{(N-1)} = \text{IP}^{(N)}(q = -1, p_k^{(N)})$, closely resemble the actual reference values whether one uses the quadratic,

$$\text{EA}_{\text{fit}} = 0.99783 \cdot \text{EA}_{\text{ref}} + 0.0012 \quad (20)$$

or the higher-order model

$$EA_{\text{fit}} = 0.99952 \cdot EA_{\text{ref}} + 0.0003 \quad (21)$$

This motivates us to use the objective function in Eq. (19) for other quantities as well. Specifically, for the properties of interest, $P = \mu, \eta, \text{IP}$, we use the weighted least-squares fitting

$$\min_{\{p_k^{(N)}\}} \sum_q \left(\frac{\mathcal{P}_{\text{ref}}^{(N)}(q) - \mathcal{P}^{(N)}(q, p_k^{(N)})}{\mathcal{P}_{\text{ref}}^{(N)}(q)(q+N)} \right)^2 \quad (22)$$

where $\mathcal{P}^{(N)}(q, p_k^{(N)})$ denotes a k-term expansion from Eqs. (14), (16), or (17).

III. Results

A. Fitting Ionization Potential (IP) and Electron Affinity (EA)

Weighted regression of the IPs for the isoelectronic series with $2 \leq N \leq 97$ gives excellent fits. For the quadratic model, the average correlation coefficient was $\langle R^2 \rangle = 0.997$ and all the correlations were excellent, with the standard deviation of the correlation coefficients being $\sigma_{R^2} = 0.0046$. The $(q+N)^{-1}$ -model gave slightly superior results, $\langle R^2 \rangle = 0.998$ and $\sigma_{R^2} = 0.0025$. It is clear that weighting the variance of the regression, Eq. (19), has a more profound effect on the quality of the fits than the length of the expansion in the interpolation model.

Figure 1 compares the electron affinity obtained by regressing the ionization potentials (and electron affinities, where available) to the reference values for the electron affinity. Results are excellent: $EA_{\text{fit}} = 0.99783 EA_{\text{ref}} + 0.0012$ ($R^2 = 1.00$) for the quadratic model and $EA_{\text{fit}} = 0.99952 EA_{\text{ref}} + 0.0003$ ($R^2 = 1.00$) for the $(q+N)^{-1}$ -model. Both models are extremely good, but the mean and standard error of the correlation coefficients is slightly better for the higher-order model ($\langle R^2 \rangle = 0.998$ and $\sigma_{R^2} = 0.0025$ vs. $\langle R^2 \rangle = 0.997$ and $\sigma_{R^2} = 0.0046$). These results correspond to fits where the first seven IPs (and EA where available) are used to determine the regression coefficients in Eq. (19). (For the corresponding first of μ and η the first six reference values were used instead.) Our tests indicate that this is the amount of data that balances

the errors due to truncating the series (14) and including data that requires higher-order correlations. In some special cases, using less data gives better results. These cases will be explicitly indicated.

B. Chemical Potential and Hardness Evaluated from Fits of IP and EA

The simplest approach to approximating the chemical potential and chemical hardness is to use the regressions from III.A to compute the IP and EA, and then compute the chemical potential and hardness directly by Eqs. (4) and (5), respectively. This did not give good results. Although weighted linear regression does an excellent job for fitting the (known) hardness values of neutral atoms (cf. Figure 2), the regressions give chemically absurd results for highly negative anions. For example, the quadratic fit predicts that almost all atoms have negative hardness. The higher-order model tends to predict that the electron affinities of anions are very positive (e.g., $EA[Cl^-] = 10.7$ eV; $EA[Cr^-] = 22.7$ eV; $EA[I^-] = 10.5$ eV). This appealing straightforward approach, therefore, does not work.

C. Chemical Hardness Evaluated from η vs. q fits

We could also estimate the chemical hardness (and chemical potential) by first computing η with all available IP's and EA's, and fitting η vs. q for each isoelectronic series using Eq. (22). It seems like this might be preferable to the method presented in the previous section, since one is not taking the difference of two different fits to the data. It is surprising, then, that Figure 3 shows that hardness of the neutral atoms estimated by direct fits are worse than those obtained in the previous section (compare to Figure 2). The errors in Figure 3 are already unpleasantly large for neutral atoms (where exact values are available): the predictions for anions are erratic.

The problem is that $\eta(q)$ is not a smooth function. To exemplify this, Figure 4 plots all the isoelectronic series for which $R^2 < 0.9$. The obvious lack of smoothness means that it is difficult (perhaps impossible) to fit $\eta(q)$ with any simple expression. This effect seems to be intrinsic to the chemical hardness. At first we thought that this effect was due to sudden changes in electron configuration (most notably for d -block and f -block elements) upon ionization, but lack of smoothness occurs also in other sectors of

the periodic table. It is likely that this problem originates from small relative errors in the IPs of highly-charged cations which results in large relative errors in the ionization-potential differences that are used to compute η . This is consistent with our observation that plots of μ vs. q are nicely regular, a feature we shall exploit in the next section.

D. Chemical Potential Evaluated from μ vs. q fits

In contrast to chemical hardness, accurate fits of μ vs. q can be obtained. As before, the higher-order model is slightly better than the quadratic model ($\langle R^2 \rangle = 0.999$ and $\sigma_{R^2} = 0.0007$ vs. $\langle R^2 \rangle = 0.999$ and $\sigma_{R^2} = 0.0014$). As it can be seen in Figure 5, both fits are essentially perfect.

Table 1 reports chemical potential of the neutral atoms, along with the values of chemical potential for anions inferred from extrapolating the quadratic model and the higher-order $(q + N)^{-1}$ -model. There is not a strong correlation between the predicted chemical potential values of negatively charged ions from the two fits, but if one focuses on just the first three periods, the two models give the same trend. (Regressing $\mu_{\text{quadratic}}$ vs. $\mu_{\text{higher-order}}$ gives a slope of 0.95 and $R^2 = 0.86$. Considering 97 atoms, the $\mu_{\text{dianion}} > \mu_{\text{anion}} > \mu_{\text{neutral}}$ condition is violated in 38 and 34 cases for the quadratic and higher-order model, respectively. However, if one chooses $\mu_{\text{anion}} = \min(\mu_{\text{quadratic}}, \mu_{\text{higher-order}})$ and $\mu_{\text{dianion}} = \max(\mu_{\text{quadratic}}, \mu_{\text{higher-order}})$, only 12 of the 97 atoms have dianions with lower chemical potentials than their anions. The violations that persist are easily understood: both models have a maximum at a negative value of q . If $-2 < q_{\text{max}} < -1$, then the chemical potential of the dianion can be less than that of the anion. The same problem can occur for the interpolation of the hardness, but in that case the maximum sometimes occurs earlier, between $q = -1$ and $q = 0$.

E. Benchmark values of Chemical Potential.

In Table 3 presents the proposed benchmark values of chemical potential. To obtain these values, we directly perform a (weighted) least-squares fit of μ vs. q to the quadratic and the higher-order models using Eq. (22). Next, we choose the chemical

potential of the anion as the lowest value predicted by the models and the chemical potential of the dianion as the highest value predicted by the models, $\mu_{\text{anion}} = \min(\mu_{\text{quadratic}}, \mu_{\text{higher-order}})$ and $\mu_{\text{dianion}} = \max(\mu_{\text{quadratic}}, \mu_{\text{higher-order}})$. After this, there are just 12 cases where the dianion has a lower chemical potential than the anion and 3 cases where the anion has a lower chemical potential than the neutral atom. For these 15 species, we use the direct fit of the ionization potentials/electron affinities described in section III.A. This leaves only three cases where $\mu_{\text{dianion}} < \mu_{\text{anion}}$ (He^{2-} , Sc^{2-} , V^{2-}). Rather than report chemically unreasonable values for these species, we choose to leave the chemical potential of these rather unimportant dianions unspecified.

F. Chemical Hardness Evaluated from μ vs. q and IP vs. q fits.

Because the results of μ vs. q and IP vs. q fits are much better than the direct η vs. q fit, it seems promising to approximate the hardness directly from these fitted quantities, using Eq. (18). The results are reported in Table 2. The results are superior to the alternative procedures considered in sections II.B and II.C, though they are still far from perfect. The biggest problem is violations of energy convexity, $\eta < 0$. However, since systems with negative hardness are unstable to disproportionation, it is not surprising that half of the dianions have negative hardness values. (Indeed, one may expect that the disproportionation reaction $2\text{A}^{2-} \rightarrow \text{A}^- + \text{A}^{3-}$ is often spontaneous (corresponding to $\eta[\text{A}^{2-}] < 0$) because one expects that (a) typically the anion is significantly more stable than the dianion and (b) the trianion is at most marginally less stable than the dianion, since resonance states for atomic trianions are seemingly unknown. The same logic can explain negative hardness values for anions, but because many atomic anions are stable, there should be (and are) many fewer anions with negative hardness values.) It is not at all surprising, therefore, that the (unbound) anions of the noble gases (group 18) are negative. Overall, 28 (24) out of the 97 anions have negative hardness when the quadratic models (higher-order models) for IP and μ are used to evaluate the hardness. In particular, for lanthanides and actinides there are many cases where $\eta \leq 0$ for the quadratic model. In these elements, the hardness seems difficult to model mainly because the extra electron

is added to a degenerate shell and so the hardness is expected to be small. (In fact, $\eta = 0$ in the frozen-orbital approximation.)

The other counterintuitive behavior we observe is that sometimes the hardness of the anion is predicted to be higher than the hardness of the neutral atom. As we discussed in section II.C.(b), this is plausible when the anion has a closed-shell electron configuration. However, we found 17 non-closed-shell cases where $\eta_{\text{anion}} > \eta_{\text{neutral}}$ for the quadratic model and 12 non-closed-shell cases for the higher-order model. Because different atoms violate this condition in the different models, we can use the same strategy used for the chemical potential and mix results from different models to improve the results for the chemical hardness.

G. Benchmark values of Chemical Hardness

Table 3 presents the benchmark values we propose for the chemical hardness. A similar approach to predicting chemical potential is taken: we try to select the models that give results in strongest agreement with chemical intuition. If superscripts are used to indicate “temporary” values of hardness, then the detailed procedure we follow for the anions is:

- 1) Choose the hardness as the maximum value from the regression models, $\eta_{\text{anion}}^{(1)} = \max(\eta_{\text{quadratic}}, \eta_{\text{higher-order}})$. This reduces the number of cases where $\eta < 0$ from 24 (from the higher-order model) to 16.
- 2) If $\eta_{\text{anion}}^{(1)} > \eta_{\text{neutral}}$, try to correct this behavior by choosing the hardness from the other model. That is, if $\eta_{\text{anion}}^{(1)} > \eta_{\text{neutral}}$, then $\eta_{\text{anion}}^{(2)} = \min(\eta_{\text{quadratic}}, \eta_{\text{higher-order}})$. This does not increase the number of cases where $\eta < 0$ but reduces the number of cases where $\eta_{\text{anion}}^{(1)} > \eta_{\text{neutral}}$ to only 6.

For the dianions:

- 3) In all cases where the hardness of the dianion is negative, choose instead a value of zero. In other cases, choose the least-positive hardness from the two models, so as to maximize the chance that $\eta_{\text{dianion}} < \eta_{\text{anion}}$. That is,

$$\eta_{\text{dianion}}^{(1)} = \begin{cases} 0 & \eta_{\text{higher-order}} \leq 0 \wedge \eta_{\text{quadratic}} \leq 0 \\ \min(\eta_{\text{quadratic}}, \eta_{\text{higher-order}}) & \eta_{\text{higher-order}} > 0 \wedge \eta_{\text{quadratic}} > 0 \end{cases} \quad (23)$$

- 4) If one still observes that $\eta_{\text{dianion}} < \eta_{\text{anion}}$, then set the hardness of the dianion to zero.

This screening removes most of the counterintuitive hardness values, but it also leaves some anions and dianions unassigned hardness values. In these cases, one may either set the hardness to zero or leave it unspecified. For anions we prefer the second approach.

While dianions are never stable in the gas phase, they are sometimes stable in solution and in crystals. It is inconvenient to leave the hardness of these chemically important dianions unspecified. For example, our method does not predict positive hardness values for the group 16 dianions (chalcogens), but O^{2-} is stable in ionic crystals like MgO , CaO , and UO_2 , and the “apparent” ionization potential of O^{2-} in an ionic crystal is a key quantity for explaining the observed trends in the heats of formation of these oxides.⁸¹

In order to evaluate μ and η for O^{2-} one must have reliable estimates for $IP^{(N=11)}(q=-3)$ and $IP^{(N=10)}(q=-2)$. However experimental IPs for the 11-electron and 10-electron atoms are available only for $q \leq 0$ and $q \leq -1$, respectively. Since none of our approaches gives a positive hardness for O^{2-} , we supplemented the set of experimental IPs with computational values. Specifically, we performed CCSD calculations for atoms with fractional atomic number, Z , but integer number of electrons, N , corresponding to the atomic charges $q = -1, -1.2, -1.4, -1.6, -1.8$, and -2.0 . For non-integer atomic charges, we used the union of the uncontracted 6-311+G basis sets of the atoms corresponding to the two closest integers to $Z = N + q$. (This alleviates the fact the Gaussian basis sets for atoms are optimized for integer Z .) As shown in Figure 6, the results from these calculations match the experimental trend, and can be added to the experimental data to perform a more reliable estimate of the ionization potential and electron affinity of O^{2-} , resulting in an estimated hardness of $\eta[\text{O}^{2-}] = 5.4 \text{ eV}$. This fits nicely with the trend of chemical hardness for the oxygen atom, $\eta[\text{O}] = 12.16 \text{ eV}$, and

oxygen anion, $\eta[\text{O}^-] = 9.61 \text{ eV}$. The corresponding value for the chemical potential of the oxygen dianion is $\mu[\text{O}^{2-}] = 9.7 \text{ eV}$.

This procedure can, in principle, be employed to provide improved estimates of the ionization potential for all the atomic trianions and dianions, along with the unbound atomic anions. We did not do this because most, if not all, of the chemically relevant atomic dianions are in group 16 (chalcogens). In the same way that the chemical potential for a few dianions were left unspecified, we believe that it is acceptable to leave the chemical hardness for some dianions unspecified as well. Also, the aforementioned computational procedure is not always robust. Computing the ionization potential becomes tricky when the critical charge (that is, the value of q for which the IP vanishes) is greater than or approximately equal to -1 , as it is for atoms with unbound or extremely weakly bound anions. (Indeed, the cases where the critical charge is close to -2 are exactly the chalcogens.) In these cases, the orbitals that host the unbound electrons in the calculation change shape as the electron becomes more and more unbound, and one would need to use a more sophisticated computational methods. Given the limited chemical relevance of the values that could be thus computed, and our preference for relying solely on experimental data whenever possible, we chose not to develop such a procedure.

H. Estimates for the Critical Charges of Atoms

The critical charge of an N -electron atom is the smallest charge, Z_{crit} , for which N electrons remain bound. This value can be found by solving the nonlinear equation,

$$\text{IP}^{(N)}(q = Z_{\text{crit}} - N) = 0 \quad (24)$$

Our models for the ionization potential of isoelectronic series (section III.A) can therefore be used to estimate the critical charge of an N -electron atom. We observe that fitting the ionization potentials to the quadratic model reproduces the known values of the critical charge more accurately, so this model is used to estimate the critical charge for $N = 2$ to $N = 96$ as reported in Table 4. The predicted Z_{crit} are typically within a few tenths of an eV of the previously reported values in the literature.^{82, 83} To the best of our

knowledge, this is the first time that ionization-potential extrapolation has been used to estimate the critical charge for isoelectronic series with more than 18 electrons.⁶⁹

IV. Summary

By fitting the asymptotic expansions of the ionization potential, Eq. (14), and the chemical potential, Eq. (16), to accurate (generally experimental) ionization potentials/electron affinities, we have provided benchmark values for the chemical potential, hardness (via Eq. (18)), and critical charge (via Eq. (24)) of atoms and atomic ions with zero (H^+ , He^{2+}) to 97 electrons (Bk). The resulting data updates, improves, and extends the previously available in the literature. (For neutral atoms, anions, and dianions the data is available in Table 3. A complete tabulation of data is available as supplementary material.) As described in sections II.E and II.G, our procedure differs from typical approaches not only because of the care we took in curating accurate data, but primarily because we use well-known chemical trends in the chemical potential and chemical hardness to select a value for the chemical potential and hardness of the anions and dianions that is in maximum agreement with chemical intuition and that preserves the regular trends of μ vs. q and η vs. q for isoelectronic series. The emphasis we place on preserving observed chemical trends is motivated by our target applications in conceptual density-functional theory, where the chemical potential and chemical hardness are used to quantify chemical trends and to apply chemical-potential/electronegativity equalization principle, hard/soft acid/base rule, and the maximum hardness rule.

We note that the proposed procedure could be extended to polyatomic functional groups and molecules. However, because the chemical potential and chemical hardness quantify changes at fixed molecular geometry, for polyatomic species the raw input data on ionization potentials and electron affinities must be obtained from computations. Developing a suitable computational protocol for the chemical potential and chemical hardness of molecules is a priority for future work.

Acknowledgments.

This work was financed by FONDECYT through project No 1140313, and also by Centers Of Excellence With Basal/Conicyt Financing, Grant FB0807,

CEDENNA. PWA acknowledges support from NSERC and Compute Canada. FHZ acknowledges support from Vanier Canada Graduate Scholarship. CC is thankful to professors Juan Botero and William Rodriguez for their hospitality during his sabbatical leave (in autumn 2015) in The Universidad Nacional de Colombia, Sede Medellin.

Figure 1. Comparison between interpolated and reference values of the electron affinity of neutral atoms with stable anions. The legend “Regular” means that a linear least-squares fit was used to regress IP vs. q , while the legend “Weights” indicates that a weighted least-squares fit was used. a) Electron affinity interpolated with a quadratic model: $a_0q^2 + a_1q + a_2$. b) Electron affinity interpolated with a higher order model: $a_0q^2 + a_1q + a_2 + a_3(q + N)^{-1}$. All values are in eV.

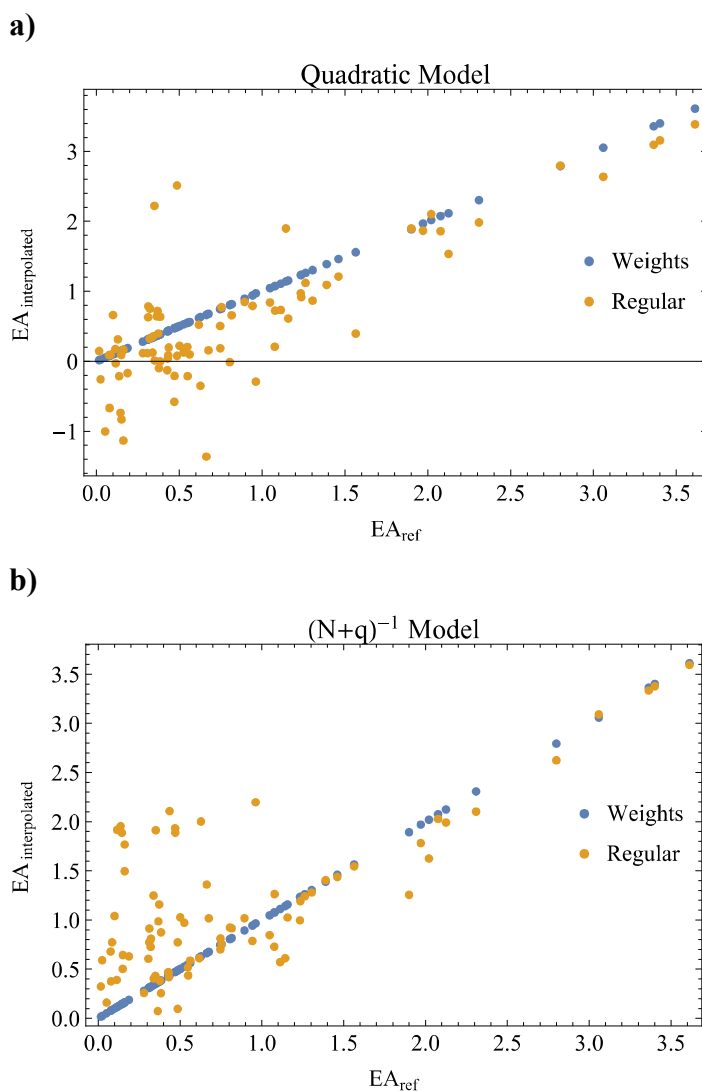
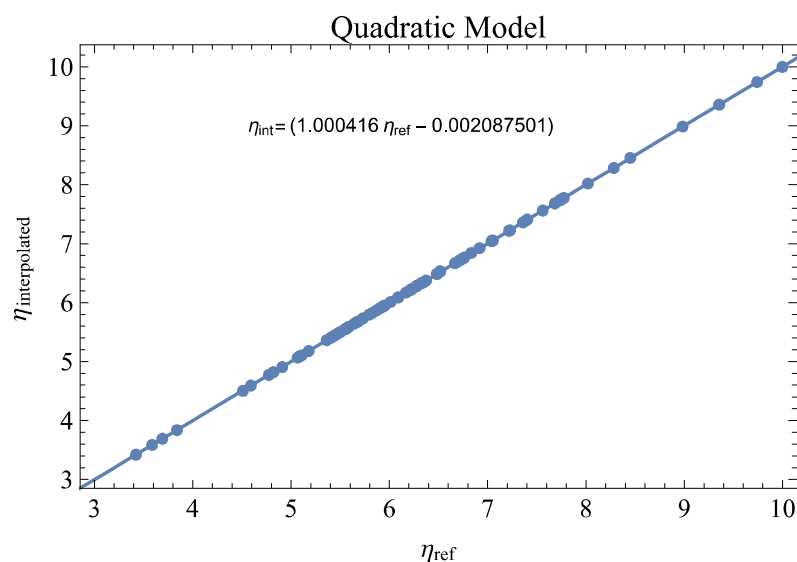


Figure 2. Comparison between interpolated and reference values of chemical hardness of neutral atoms with stable anions. Regression of the IP and EA were first made, and then the chemical hardness was computed from Equation (5). a) Ionization potential and electron affinity were fitted to a quadratic model: $a_0q^2 + a_1q + a_2$. b) Ionization potential and electron affinity were fitted to a higher order model: $a_0q^2 + a_1q + a_2 + a_3(q + N)^{-1}$. All values are in eV.

a)



b)

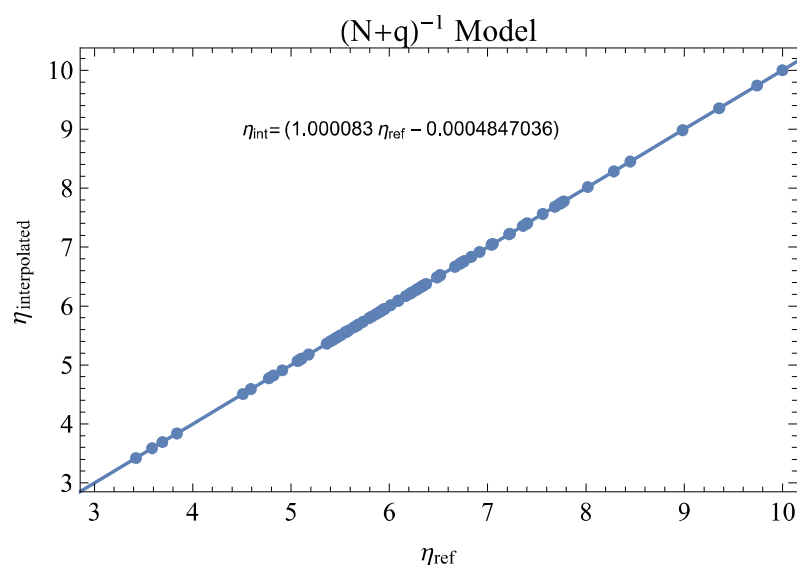
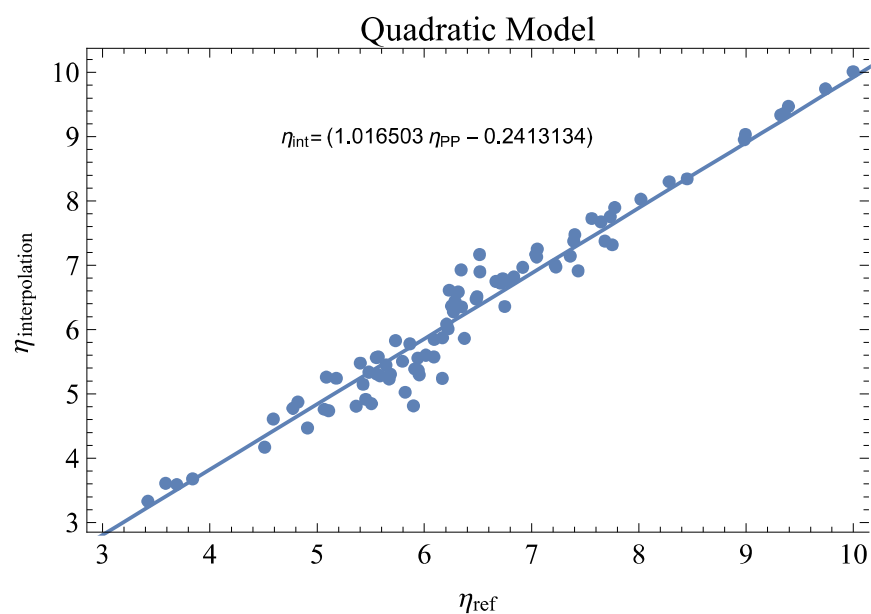


Figure 3. Comparison between interpolated and reference values of chemical hardness of neutral atoms with stable anions. Hardness of the neutral atoms is estimated by direct fits to the reference hardness. a) Fitting to a quadratic model: $a_0q^2 + a_1q + a_2$. b) Fitting to a higher order model: $a_0q^2 + a_1q + a_2 + a_3(q + N)^{-1}$. All values are in eV.

a)



b)

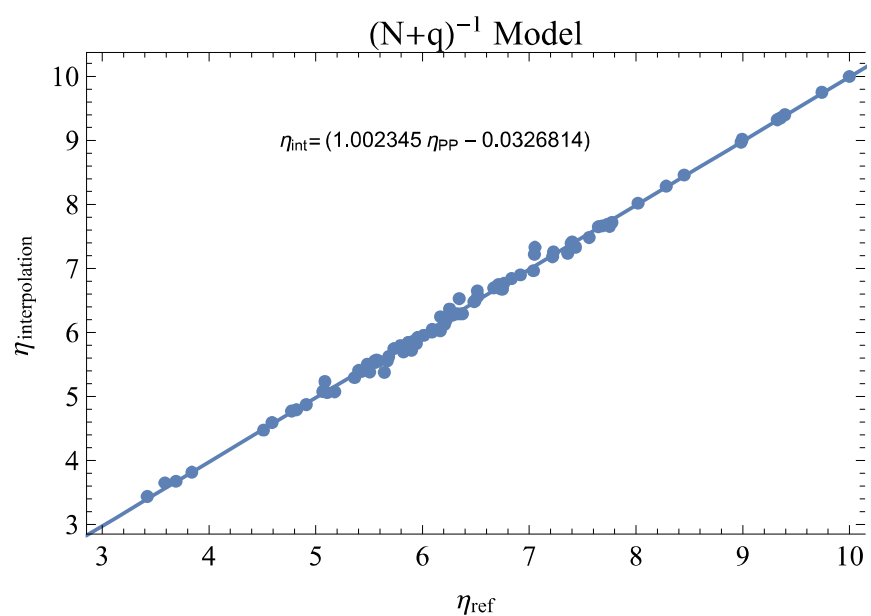


Figure 4. Plots of the hardness vs. charge of the isoelectronic for which the coefficient of correlation of the higher order model is $R^2 < 0.9$. Note the lack of smoothness of the data for these series, which make them very difficult, if possible, to fit to a simple expression.

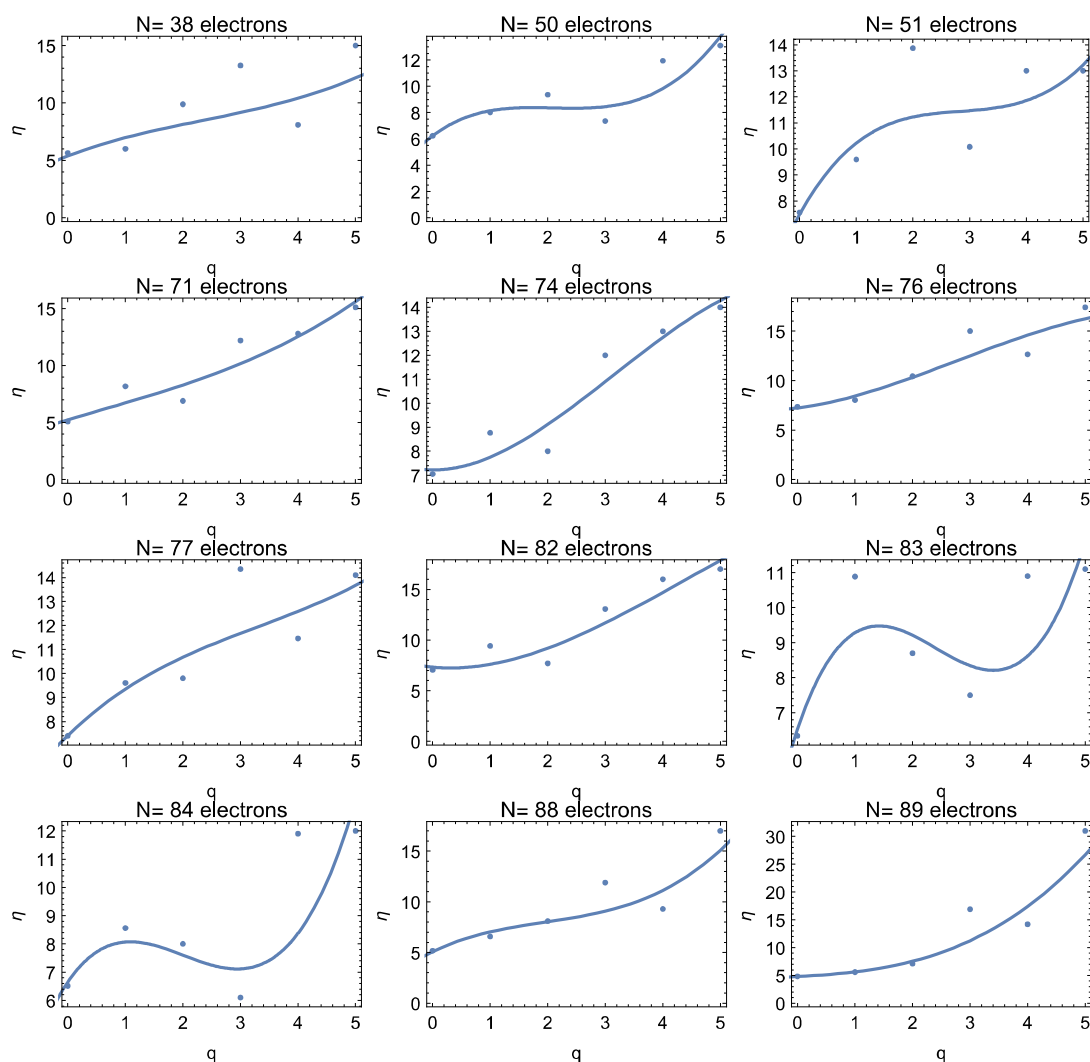
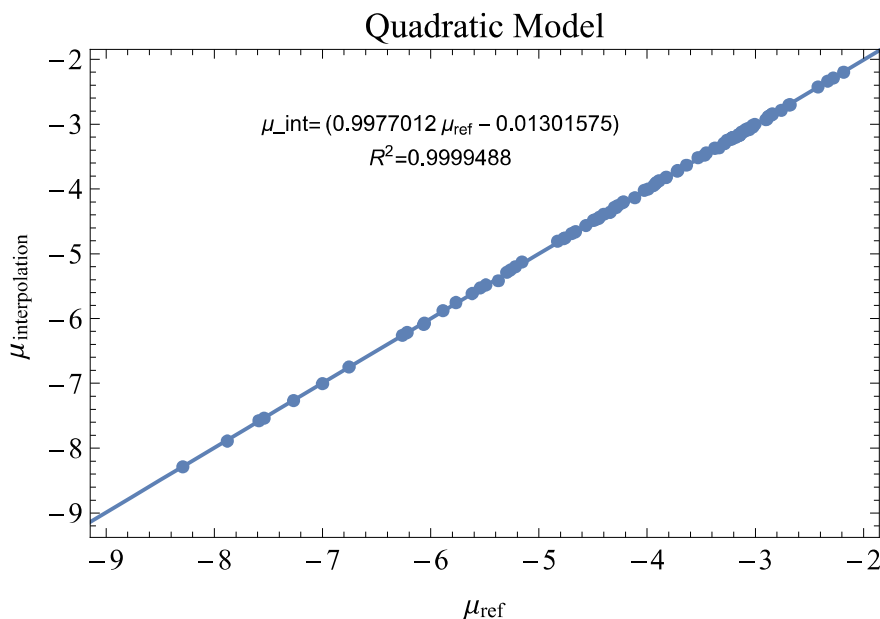


Figure 5. Comparison between interpolated and reference values of the chemical potential of neutral atoms with stable anions. Chemical potential of the neutral atoms is estimated by direct fits to the reference chemical potential. a) Fitting to a quadratic model: $a_0 q^2 + a_1 q + a_2$. b) Fitting to a higher order model: $a_0 q^2 + a_1 q + a_2 + a_3 (q + N)^{-1}$. All values are in eV.

a)



b)

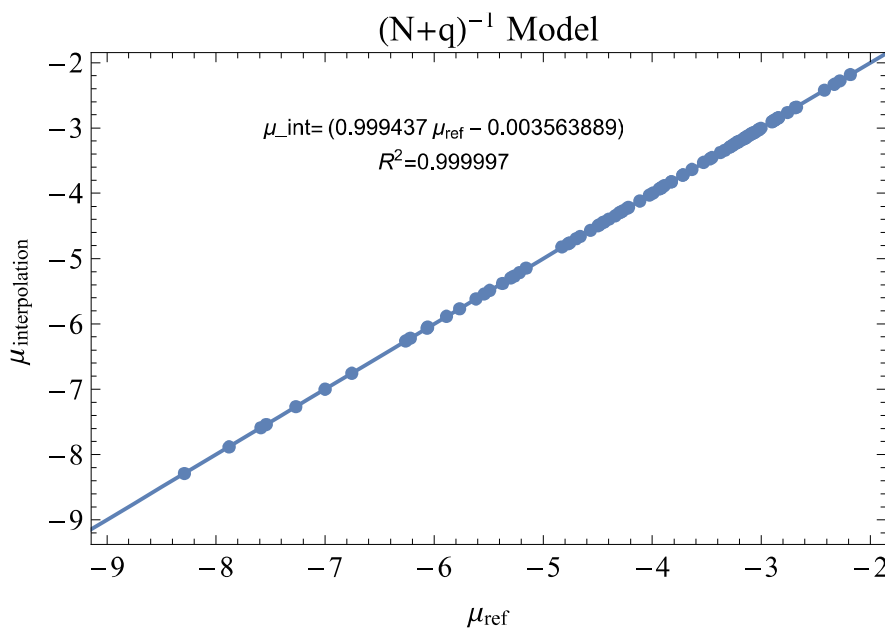
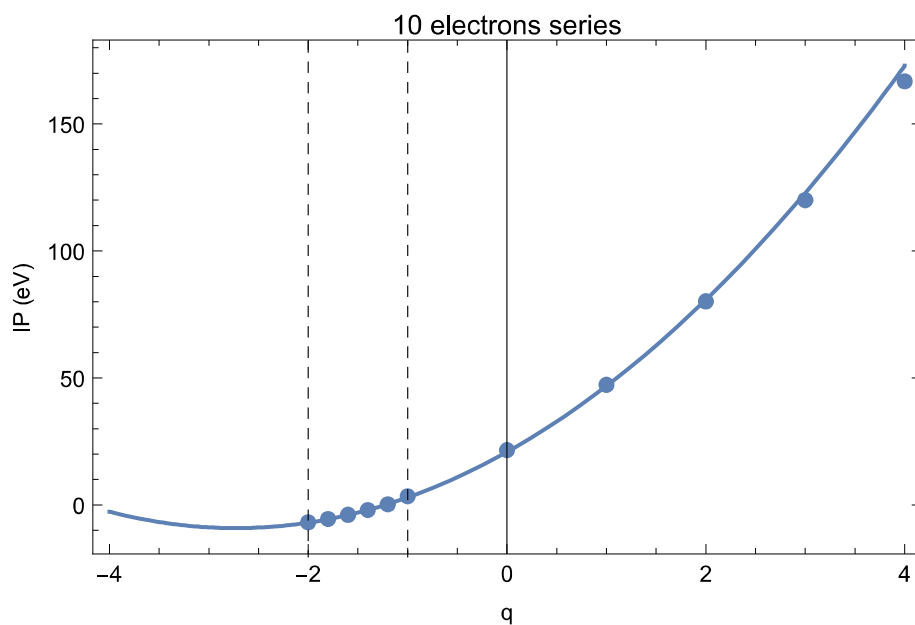


Figure 6. Ionization potential of a) the 10-electron series and b) the 11-electron series. The points between dashed vertical lines come from CCSD calculations (see text for details) while points for $q \geq 0$ are experimental data. The solid line is the quadratic polynomial that fits the data.

a)



b)

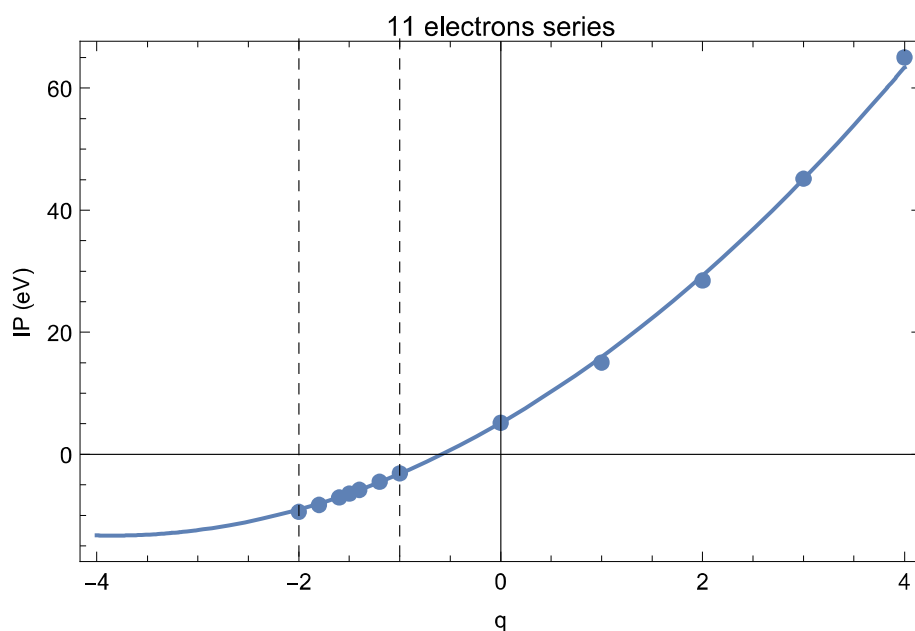


Table 1. Chemical potential of neutral atoms and negatively charged ions. For neutral atoms with known electron affinity the values have been determined by interpolation, while for those atoms with unknown electron affinity the reported values were extrapolated from the corresponding isoelectronic series. For anions and dianions all values were determined by extrapolation of the μ vs. q isoelectronic series, as explained in the text. Columns labeled q^2 correspond to a quadratic regression model while columns labeled $(q+N)^{-1}$ corresponds to a regression model that includes the next term after the quadratic one in the expansion of Eq (16). Column Ref. denotes the reference chemical potential of neutral atoms. All values are in eV.

	Ref.	$(q+N)^{-1}$			q^2		
		neutral	anion	dianion	neutral	anion	dianion
Hydrogen	-7.18	-7.18	-2.46	-5.47	-7.18	-1.24	-2.28
Helium	-12.29	-12.29	0.38	-7.15	-12.29	0.83	-1.79
Lithium	-3.00	-3.00	-0.69	-0.76	-3.00	0.33	1.72
Beryllium	-4.66	-4.66	1.70	2.31	-4.66	2.23	4.36
Boron	-4.29	-4.29	2.08	2.87	-4.29	2.56	5.62
Carbon	-6.26	-6.26	2.08	5.98	-6.26	2.75	8.41
Nitrogen	-7.27	-7.27	3.40	7.81	-7.27	4.01	9.95
Oxygen	-7.54	-7.54	2.79	-0.65	-7.54	3.34	3.87
Fluorine	-10.41	-10.41	-1.91	0.87	-10.41	-0.71	3.70
Neon	-10.78	-10.78	1.44	-0.41	-10.77	2.19	2.77
Sodium	-2.84	-2.84	0.41	2.60	-2.84	1.26	4.51
Magnesium	-3.82	-3.82	1.88	3.92	-3.82	2.39	5.58
Aluminum	-3.21	-3.21	1.69	4.39	-3.21	2.15	6.44
Silicon	-4.77	-4.77	1.59	5.90	-4.77	2.16	7.97
Phosphorus	-5.62	-5.62	2.01	6.68	-5.61	2.59	8.44
Sulfur	-6.22	-6.22	1.28	3.59	-6.22	1.78	0.37
Chlorine	-8.29	-8.29	-0.90	2.51	-8.29	-1.82	-3.43
Argon	-7.88	-7.88	1.00	-4.20	-7.89	-0.65	-4.78
Potassium	-2.42	-2.42	-1.30	-8.02	-2.43	-1.47	-4.98
Calcium	-3.07	-3.07	-2.48	-11.78	-3.07	-1.63	-4.50
Scandium	-3.37	-3.38	-3.38	-4.46	-3.37	-1.33	-2.26
Titanium	-3.45	-3.45	-1.12	-2.15	-3.45	-0.50	-3.46
Vanadium	-3.64	-3.64	-0.71	-5.56	-3.63	-1.08	-2.48
Chromium	-3.72	-3.72	-1.40	-8.97	-3.73	-0.52	-3.14
Manganese	-3.72	-3.72	-2.53	-13.82	-3.72	-0.87	-3.97
Iron	-4.03	-4.03	-4.00	-4.10	-4.02	-1.19	2.40

Cobalt	-4.27	-4.27	-0.72	4.43	-4.26	1.15	6.73
Nickel	-4.40	-4.40	1.68	0.68	-4.39	2.36	4.80
Copper	-4.48	-4.48	0.18	5.40	-4.48	1.39	6.88
Zinc	-4.70	-4.70	2.65	5.61	-4.69	3.08	6.88
Gallium	-3.21	-3.21	2.07	2.46	-3.21	2.45	8.03
Germanium	-4.57	-4.57	0.97	6.07	-4.57	2.62	8.76
Arsenic	-5.30	-5.30	1.87	5.05	-5.29	2.66	9.18
Selenium	-5.89	-5.88	0.70	3.68	-5.88	1.94	2.66
Bromine	-7.59	-7.59	-0.58	4.26	-7.58	-0.88	0.27
Krypton	-7.00	-7.00	1.63	-1.78	-7.00	0.47	-0.27
Rubidium	-2.33	-2.33	-0.37	-8.35	-2.34	0.07	0.43
Strontium	-2.87	-2.88	-2.34	-5.14	-2.88	0.22	0.56
Yttrium	-3.26	-3.27	-1.46	1.73	-3.25	0.21	2.95
Zirconium	-3.53	-3.53	0.69	-2.42	-3.52	1.05	1.35
Niobium	-3.83	-3.83	-0.76	-3.02	-3.82	0.35	3.14
Molybdenum	-3.92	-3.92	-0.56	-6.00	-3.91	1.25	3.98
Technetium	-3.92	-3.91	-1.42	-1.92	-3.90	1.52	6.70
Ruthenium	-4.22	-4.22	0.17	15.66	-4.20	2.71	8.25
Rhodium	-4.30	-4.30	5.31	5.36	-4.29	3.11	7.30
Palladium	-4.45	-4.44	1.75	1.86	-4.46	2.34	5.75
Silver	-4.44	-4.44	0.43	5.23	-4.44	1.59	7.54
Cadmium	-4.50	-4.50	2.44	0.94	-4.49	3.12	8.34
Indium	-3.09	-3.09	0.65	4.69	-3.08	2.86	10.25
Tin	-4.23	-4.23	1.69	7.33	-4.21	3.36	9.03
Antimony	-4.83	-4.82	2.01	6.25	-4.81	2.52	8.35
Tellurium	-5.49	-5.49	0.95	6.69	-5.48	1.59	0.18
Iodine	-6.76	-6.75	0.36	8.35	-6.75	-1.62	-4.93
Xenon	-6.06	-6.07	2.40	7.86	-6.09	-1.47	-5.72
Cesium	-2.18	-2.18	1.94	2.34	-2.20	-2.04	-7.91
Barium	-2.68	-2.68	-0.10	-2.88	-2.70	-3.10	-8.42
Lanthanum	-3.02	-3.03	-1.70	-3.78	-3.05	-3.32	-9.05
Cerium	-3.08	-3.09	-2.09	-3.62	-3.10	-3.63	-7.49
Praseodymium	-3.22	-3.22	-1.55	-2.16	-3.23	-2.68	-6.18
Neodymium	-2.84	-2.85	-1.00	8.44	-2.85	-2.18	-7.18
Promethium	-2.86	-2.86	1.88	12.44	-2.86	-2.70	-5.92
Samarium	-2.90	-2.91	3.28	4.03	-2.93	-2.12	-7.99
Europium	-2.89	-2.90	0.44	5.25	-2.92	-3.08	-7.82
Gadolinium	-3.14	-3.15	0.83	5.38	-3.17	-3.01	-7.69
Terbium	-3.15	-3.15	0.90	2.71	-3.18	-2.94	-7.72

Dysprosium	-3.15	-3.15	0.15	-4.19	-3.17	-2.92	-3.29
Holmium	-3.18	-3.18	-1.54	3.86	-3.20	-1.28	1.62
Erbium	-3.21	-3.21	1.15	2.89	-3.21	0.47	2.95
Thulium	-3.10	-3.10	1.08	11.69	-3.10	1.10	6.51
Ytterbium	-3.13	-3.13	4.37	12.57	-3.13	2.81	6.68
Lutetium	-2.88	-2.88	4.37	10.02	-2.89	2.59	5.39
Hafnium	-3.47	-3.47	3.19	8.74	-3.48	1.79	3.16
Tantalum	-3.94	-3.94	2.33	9.13	-3.95	0.64	5.98
Tungsten	-4.34	-4.35	3.08	1.50	-4.36	2.14	3.79
Rhenium	-3.99	-4.00	0.15	-2.19	-4.00	0.84	3.38
Osmium	-4.76	-4.76	-1.18	2.46	-4.76	0.50	7.03
Iridium	-5.27	-5.27	0.39	3.75	-5.25	1.78	8.22
Platinum	-5.54	-5.54	0.53	2.97	-5.53	1.90	6.96
Gold	-5.77	-5.77	0.40	7.77	-5.75	1.62	8.28
Mercury	-5.22	-5.22	3.13	4.82	-5.20	3.28	9.40
Thallium	-3.24	-3.24	2.00	21.29	-3.24	3.38	11.84
Lead	-3.89	-3.88	7.11	2.93	-3.88	4.22	9.53
Bismuth	-4.11	-4.12	0.74	17.99	-4.14	2.76	11.97
Polonium	-5.16	-5.15	5.13	13.46	-5.13	3.27	4.36
Astatine	-6.06	-6.05	3.05	6.26	-6.07	0.24	0.94
Radon	-5.37	-5.38	2.10	8.34	-5.42	0.51	0.60
Francium	-2.28	-2.28	2.53	15.21	-2.29	0.22	-0.40
Radium	-2.69	-2.69	4.54	6.80	-2.70	-0.14	-2.31
Actinium	-2.76	-2.76	1.38	0.24	-2.79	-1.36	-2.83
Thorium	-3.34	-3.35	-0.46	2.53	-3.37	-1.38	-2.78
Protactinium	-3.14	-3.14	0.20	-1.24	-3.15	-1.39	-3.38
Uranium	-3.28	-3.29	-0.97	6.41	-3.30	-1.60	-2.96
Neptunium	-3.29	-3.29	1.46	-7.42	-3.30	-1.35	-4.33
Plutonium	-3.06	-3.07	-2.70	0.22	-3.08	-1.78	-2.10
Americium	-3.02	-3.03	-0.24	0.66	-3.02	-0.93	-2.23
Curium	-3.16	-3.16	-0.06	1.27	-3.17	-0.92	-3.58
Berkelium	-3.11	-3.12	-0.01		-3.12	-1.46	

Table 2. Chemical hardness of neutral atoms and negatively charged ions evaluated from fitted values of chemical potential and IP's: $\eta = 2(\text{IP} + \mu)$. Chemical potentials were taken from Table 1. For neutral atoms with known electron affinity the values have been assigned by interpolation while for atoms with unknown electron affinity, the ionization potential was extrapolated from the corresponding isoelectronic series. Columns labeled q^2 correspond to a quadratic regression model while columns labeled $(q+N)^{-1}$ corresponds to a regression model that includes the next term after the quadratic one in the expansion of Eq (17). Column Ref. denotes the reference chemical hardness of neutral atoms. All values are in eV.

		$(q+N)^{-1}$			q^2	
	Ref.	neutral	anion	dianion	neutral	anion
Hydrogen	12.84	-3.42	-7.90	12.84	-0.97	-3.88
Helium	24.59	-0.04	-15.14	24.59	0.52	-5.82
Lithium	4.77	-0.15	-5.99	4.78	1.90	-2.85
Beryllium	9.32	2.04	-1.05	9.33	2.70	1.45
Boron	8.02	4.72	-2.90	8.02	5.68	1.26
Carbon	10.00	6.68	-1.55	10.00	8.02	0.44
Nitrogen	14.53	5.99	2.62	14.53	6.46	5.17
Oxygen	12.16	8.50	-14.97	12.16	9.61	-7.56
Fluorine	14.02	2.98	-4.56	14.03	5.39	-2.22
Neon	21.57	0.81	-4.88	21.59	1.39	-0.57
Sodium	4.59	1.91	-1.68	4.60	3.63	-0.86
Magnesium	7.65	2.34	1.80	7.65	2.52	3.48
Aluminum	5.55	4.24	1.46	5.55	5.16	4.37
Silicon	6.76	5.97	2.27	6.76	7.11	5.01
Phosphorus	9.74	5.51	3.78	9.75	6.67	5.99
Sulfur	8.28	6.71	-2.05	8.28	7.71	-9.68
Chlorine	9.36	5.43	26.54	9.36	3.58	-0.42
Argon	15.76	9.12	-14.89	15.74	1.44	-10.28
Potassium	3.84	-1.60	-25.93	3.82	-1.93	-13.42
Calcium	6.09	-4.91	-31.69	6.09	-3.20	-12.28
Scandium	6.37	-6.39	-13.47	6.38	-2.29	-8.15
Titanium	6.75	-2.08	-9.21	6.76	-0.84	-8.95
Vanadium	6.22	-0.36	-16.26	6.23	-1.11	-7.05
Chromium	6.09	-1.44	27.60	6.07	0.31	1.57
Manganese	7.43	12.51	-34.20	7.43	4.51	-13.11
Iron	7.75	-7.70	-6.30	7.76	-2.08	3.01
Cobalt	7.22	-0.11	5.97	7.24	3.64	7.55
Nickel	6.48	5.67	-6.10	6.50	7.02	0.55

Copper	6.49	2.83	-0.91	6.49	5.24	-0.94
Zinc	9.40	2.62	3.49	9.41	2.56	4.60
Gallium	5.57	5.00	-2.98	5.58	5.75	6.35
Germanium	6.66	4.42	3.26	6.66	7.70	5.68
Arsenic	8.99	5.35	0.40	9.00	6.93	7.55
Selenium	7.74	5.45	-1.58	7.74	7.91	-4.95
Bromine	8.45	5.57	13.31	8.47	4.95	-0.66
Krypton	14.00	5.68	-8.36	14.00	1.52	-3.81
Rubidium	3.69	0.23	-23.29	3.67	1.12	-5.06
Strontium	5.63	-4.58	-17.79	5.63	0.55	-4.63
Yttrium	5.90	-2.31	-0.80	5.93	1.03	0.00
Zirconium	6.21	2.22	-10.07	6.23	2.94	-1.69
Niobium	5.86	0.27	-9.94	5.88	2.49	0.80
Molybdenum	6.35	0.37	-15.22	6.36	4.00	1.37
Technetium	6.41	-1.74	-2.78	6.44	4.13	8.82
Ruthenium	6.29	2.48	28.10	6.32	7.56	11.46
Rhodium	6.32	12.90	2.94	6.34	8.49	5.11
Palladium	7.78	4.63	-4.05	7.75	5.80	1.95
Silver	6.27	3.46	-1.16	6.27	5.79	0.59
Cadmium	8.99	2.70	-6.09	9.01	3.17	6.69
Indium	5.40	2.06	2.56	5.41	6.50	10.06
Tin	6.23	5.59	5.40	6.27	8.93	6.97
Antimony	7.58	6.11	3.61	7.60	7.13	6.71
Tellurium	7.05	5.83	5.82	7.06	7.11	-8.90
Iodine	7.39	6.84	37.62	7.40	2.87	1.73
Xenon	12.13	12.76	9.65	12.08	2.16	-11.69
Cesium	3.42	4.82	-4.02	3.39	-3.14	-17.51
Barium	5.06	0.10	-11.91	5.02	-5.90	-16.81
Lanthanum	5.10	-2.47	-12.44	5.05	-5.70	-17.25
Cerium	4.90	-2.92	-9.29	4.88	-5.99	-12.62
Praseodymium	4.50	-1.17	-10.42	4.49	-3.42	-13.37
Neodymium	5.36	-1.68	5.47	5.35	-4.03	-15.83
Promethium	5.45	4.03	15.15	5.44	-5.13	-13.46
Samarium	5.48	6.89	-3.54	5.43	-3.91	-17.86
Europium	5.54	1.11	0.49	5.50	-5.94	-17.11
Gadolinium	6.01	1.92	2.10	5.96	-5.74	-15.72
Terbium	5.42	2.67	-3.19	5.37	-5.00	-16.24
Dysprosium	5.58	1.00	-16.18	5.54	-5.13	-7.56
Holmium	5.68	-2.41	-2.14	5.64	-1.88	-2.71
Erbium	5.79	2.92	-4.13	5.80	1.57	-2.16
Thulium	6.17	2.19	0.45	6.17	2.23	2.05
Ytterbium	6.26	2.84	17.10	6.25	3.53	5.15

Lutetium	5.08	9.42	8.31	5.07	5.87	0.74
Hafnium	6.71	6.60	3.87	6.69	3.81	-3.44
Tantalum	7.23	5.30	10.42	7.20	1.92	4.32
Tungsten	7.04	7.80	-10.35	7.01	5.91	-3.53
Rhenium	7.67	0.60	-13.31	7.67	1.98	-1.88
Osmium	7.35	-0.20	2.34	7.36	3.16	7.05
Iridium	7.40	3.92	2.86	7.43	6.68	8.96
Platinum	6.83	5.32	-1.57	6.86	8.04	4.47
Gold	6.91	5.41	-1.77	6.95	7.84	1.33
Mercury	10.44	2.73	0.92	10.48	3.71	8.25
Thallium	5.73	4.74	32.68	5.74	7.52	12.01
Lead	7.06	14.96	-3.56	7.07	9.17	8.21
Bismuth	6.34	3.37	29.23	6.29	7.40	13.76
Polonium	6.53	14.04	19.52	6.57	10.31	0.16
Astatine	6.54	11.68	-4.49	6.50	6.06	7.49
Radon	10.73	0.89	8.53	10.66	4.80	-1.84
Francium	3.58	6.04	21.35	3.57	1.41	-4.27
Radium	5.18	9.27	0.62	5.16	-0.08	-9.23
Actinium	5.23	3.47	-8.91	5.18	-2.02	-8.92
Thorium	5.92	-0.18	-4.19	5.87	-2.02	-9.73
Protactinium	5.49	1.18	-11.83	5.48	-2.00	-10.01
Uranium	5.81	-1.18	2.76	5.79	-2.46	-10.02
Neptunium	5.95	3.55	-24.84	5.93	-2.07	-11.94
Plutonium	5.92	-5.23	-8.80	5.89	-3.40	-8.91
Americium	5.89	-0.32	-6.92	5.91	-1.70	-8.58
Curium	5.66	0.52		5.64	-1.21	

Table 3. This Table summarizes the proposed benchmark values of chemical hardness and chemical potential of neutral atoms and negatively charged ions. Cases for which a value could not be assigned are marked with “-”. All values are in eV.

Charge	Chemical potential, μ .			Chemical hardness, η .		
	Neutral	Anion	Dianion	Neutral	Anion	Dianion
Hydrogen	-7.18	-2.46	-2.28	12.84	-	0.0
Helium	-12.29	0.38	-	24.59	0.52	0.0
Lithium	-3.00	-0.69	1.72	4.77	1.90	0.0
Beryllium	-4.66	1.70	4.36	9.32	2.70	0.0
Boron	-4.29	2.08	5.62	8.02	5.68	0.0
Carbon	-6.26	2.08	8.41	10.00	8.02	0.0
Nitrogen	-7.27	3.40	9.95	14.53	6.46	2.6
Oxygen	-7.54	2.79	3.87	12.16	9.61	5.4
Fluorine	-10.41	-1.91	3.70	14.02	5.39	0.0
Neon	-10.78	1.44	2.77	21.56	1.39	0.0
Sodium	-2.84	0.41	4.51	4.59	3.63	0.0
Magnesium	-3.82	1.88	5.58	7.65	2.52	1.8
Aluminum	-3.21	1.69	6.44	5.55	5.16	1.5
Silicon	-4.77	1.59	7.97	6.76	5.97	2.3
Phosphorus	-5.62	2.01	8.44	9.74	6.67	3.8
Sulfur	-6.22	1.28	3.59	8.28	7.71	0.0
Chlorine	-8.29	-1.82	2.51	9.35	5.43	0.0
Argon	-7.88	-0.65	-0.11	15.76	9.12	0.0
Potassium	-2.42	-1.47	0.49	3.84	-	0.0
Calcium	-3.07	-2.48	0.39	6.09	-	0.0
Scandium	-3.38	0.81	-	6.37	-	0.0
Titanium	-3.45	-1.12	-0.39	6.75	-	0.0
Vanadium	-3.64	-1.08	-	6.22	-	0.0
Chromium	-3.72	-2.30	-1.64	6.09	0.31	0.0
Manganese	-3.72	-2.53	-0.24	7.43	4.51	0.0
Iron	-4.03	-4.00	2.40	7.75	-	0.0
Cobalt	-4.27	-0.72	6.73	7.22	3.64	6.0
Nickel	-4.40	1.68	4.80	6.48	5.67	0.0
Copper	-4.48	0.18	6.88	6.49	5.24	0.0
Zinc	-4.70	2.65	6.88	9.39	2.62	0.0
Gallium	-3.21	2.07	8.03	5.57	5.00	0.0
Germanium	-4.57	0.97	8.76	6.67	4.42	3.3
Arsenic	-5.30	1.87	9.18	8.98	6.93	0.4

Selenium	-5.88	0.70	3.68	7.73	5.45	0.0
Bromine	-7.59	-0.88	4.26	8.45	5.57	0.0
Krypton	-7.00	0.47	2.41	14.00	5.68	0.0
Rubidium	-2.33	-0.37	0.43	3.69	1.12	0.0
Strontium	-2.88	-2.34	0.56	5.64	0.55	0.0
Yttrium	-3.27	-1.46	2.95	5.91	1.03	0.0
Zirconium	-3.53	0.69	1.35	6.21	2.94	0.0
Niobium	-3.83	-0.76	3.14	5.86	2.49	0.0
Molybdenum	-3.92	-0.56	3.98	6.35	4.00	0.0
Technetium	-3.91	-1.42	6.70	6.73	4.13	0.0
Ruthenium	-4.22	0.17	15.66	6.28	2.48	0.0
Rhodium	-4.30	3.11	7.30	6.32	-	2.9
Palladium	-4.44	1.75	5.75	7.77	5.80	0.0
Silver	-4.44	0.43	7.54	6.27	5.79	0.0
Cadmium	-4.50	2.44	8.34	8.99	3.17	0.0
Indium	-3.09	0.65	10.25	5.40	2.06	2.6
Tin	-4.23	1.69	9.03	6.23	5.59	5.4
Antimony	-4.82	2.01	8.35	7.56	7.13	3.6
Tellurium	-5.49	0.95	6.69	7.04	5.83	0.0
Iodine	-6.75	-1.62	8.35	7.39	6.84	1.7
Xenon	-6.07	-1.47	7.86	12.13	2.16	0.0
Cesium	-2.18	-2.04	2.34	3.42	-	0.0
Barium	-2.68	-3.10	-2.88	5.07	0.10	0.0
Lanthanum	-3.03	-0.45	2.11	5.11	-	0.0
Cerium	-3.09	-3.63	-3.62	4.91	-	0.0
Praseodymium	-3.22	-2.68	-2.16	4.51	-	0.0
Neodymium	-2.85	-2.18	8.44	5.36	-	0.0
Promethium	-2.86	-2.70	12.44	5.45	4.03	0.0
Samarium	-2.91	-2.12	4.03	5.48	-	0.0
Europium	-2.90	-3.08	5.25	5.55	1.11	0.0
Gadolinium	-3.15	-3.01	5.38	6.01	1.92	0.0
Terbium	-3.15	-2.94	2.71	5.43	2.67	0.0
Dysprosium	-3.15	0.07	2.35	5.59	1.00	0.0
Holmium	-3.18	-1.54	3.86	5.68	-	0.0
Erbium	-3.21	0.47	2.95	5.80	2.92	0.0
Thulium	-3.10	1.08	11.69	6.17	2.23	0.5
Ytterbium	-3.13	2.81	12.57	6.25	3.53	0.0
Lutetium	-2.88	2.59	10.02	5.09	-	0.7
Hafnium	-3.47	1.79	8.74	6.71	6.60	0.0

Tantalum	-3.94	0.64	9.13	7.23	5.30	4.3
Tungsten	-4.35	2.14	3.79	7.05	5.91	0.0
Rhenium	-4.00	0.15	3.38	7.68	1.98	0.0
Osmium	-4.76	-1.18	7.03	7.36	3.16	2.3
Iridium	-5.27	0.39	8.22	7.40	6.68	2.9
Platinum	-5.54	0.53	6.96	6.83	5.32	0.0
Gold	-5.77	0.40	8.28	6.92	5.41	0.0
Mercury	-5.22	3.13	9.40	10.44	3.71	0.9
Thallium	-3.24	2.00	21.29	5.73	4.74	0.0
Lead	-3.88	4.22	9.53	7.05	-	0.0
Bismuth	-4.12	0.74	17.99	6.34	3.37	0.0
Polonium	-5.15	3.27	13.46	6.51	-	0.0
Astatine	-6.05	0.24	6.26	6.52	6.06	0.0
Radon	-5.38	0.51	8.34	10.75	4.80	0.0
Francium	-2.28	0.22	15.21	3.59	1.41	0.0
Radium	-2.69	-0.14	6.80	5.18	0.00	0.0
Actinium	-2.76	-1.36	0.24	4.82	3.47	0.0
Thorium	-3.35	-1.38	2.53	5.94	-	0.0
Protactinium	-3.14	-1.39	-1.24	5.51	1.18	0.0
Uranium	-3.29	-1.60	6.41	5.82	-	0.0
Neptunium	-3.29	-1.35	1.83	5.95	-	0.0
Plutonium	-3.07	-2.70	0.22	5.94	-	0.0
Americium	-3.03	-0.93	0.66	5.90	-	0.0
Curium	-3.16	-0.92	1.27	5.67	0.52	-
Berkelium	-3.12	-	-	6.17	-	-

Published on 19 August 2016. Downloaded by Northern Illinois University on 19/08/2016 13:20:34.

Table 4. Critical nuclear charge (Z critical) of the isoelectronic series with $2 \leq N \leq 96$. All values were assigned using a quadratic model fitted to seven points with weighted variance objective function except for: cases marked with “*” ($N = 19, 26, 55, 59, 60, 87$) for which equal weights were used and, cases the marked with “**” ($N = 87$), for which four points were used in the fitting.

N	Isoelectronic Atom	Z critical	N	Isoelectronic Atom	Z critical	N	Isoelectronic Atom	Z critical
2	Helium	0.917	34	Selenium	32.897	66	Dysprosium	64.798
3	Lithium	2.181	35	Bromine	33.759	67	Holmium	65.855
4	Beryllium	2.871	36	Krypton	34.636	68	Erbium	66.868
5	Boron	4.142	37	Rubidium	35.867	69	Thulium	67.926
6	Carbon	4.962	38	Strontium	36.853	70	Ytterbium	68.997
7	Nitrogen	5.864	39	Yttrium	37.988	71	Lutetium	70.184
8	Oxygen	7.070	40	Zirconium	38.932	72	Hafnium	70.938
9	Fluorine	7.878	41	Niobium	39.910	73	Tantalum	71.981
10	Neon	8.753	42	Molybdenum	40.789	74	Tungsten	72.947
11	Sodium	10.276	43	Technetium	41.844	75	Rhenium	73.855
12	Magnesium	10.895	44	Ruthenium	42.896	76	Osmium	74.978
13	Aluminum	12.196	45	Rhodium	43.773	77	Iridium	75.831
14	Silicon	12.927	46	Palladium	44.780	78	Platinum	76.744
15	Phosphorus	13.803	47	Silver	45.909	79	Gold	77.668
16	Sulfur	14.904	48	Cadmium	46.808	80	Mercury	78.693
17	Chlorine	15.765	49	Indium	48.230	81	Thallium	80.204
18	Argon	16.636	50	Tin	48.938	82	Lead	80.941
19	Potassium	16.674*	51	Antimony	49.842	83	Bismuth	81.947
20	Calcium	18.800	52	Tellurium	50.856	84	Polonium	82.864
21	Scandium	19.993	53	Iodine	51.740	85	Astatine	83.747
22	Titanium	20.951	54	Xenon	52.626	86	Radon	84.627
23	Vanadium	21.981	55	Cesium	54.623*	87	Francium	85.835**
24	Chromium	22.845	56	Barium	54.752	88	Radium	86.840
25	Manganese	23.811	57	Lanthanum	55.945	89	Actinium	87.968
26	Iron	25.238*	58	Cerium	56.741	90	Thorium	88.906
27	Cobalt	25.970	59	Praseodymium	58.113*	91	Protactinium	89.882
28	Nickel	26.828	60	Neodymium	59.088*	92	Uranium	90.893
29	Copper	27.777	61	Promethium	59.935	93	Neptunium	91.885
30	Zinc	28.818	62	Samarium	60.950	94	Plutonium	92.911
31	Gallium	30.258	63	Europium	61.940	95	Americium	93.973
32	Germanium	30.931	64	Gadolinium	62.959	96	Curium	94.980

33	Arsenic	31.828	65	Terbium	63.949			
----	---------	--------	----	---------	--------	--	--	--

Published on 19 August 2016. Downloaded by Northern Illinois University on 19/08/2016 13:20:34.

Physical Chemistry Chemical Physics Accepted Manuscript

References:

1. R. G. Parr, R. A. Donnelly, M. Levy and W. E. Palke, *J Chem Phys*, 1978, 68, 3801-3807.
2. R. G. Parr and R. G. Pearson, *J Am Chem Soc*, 1983, 105, 7512-7516.
3. R. P. Itzkowski and J. L. Margrave, *J Am Chem Soc*, 1961, 83, 3547-3551.
4. R. T. Sanderson, *Science*, 1951, 114, 670-672.
5. R. G. Pearson, *Science*, 1966, 151, 172-177.
6. R. G. Pearson, *J Am Chem Soc*, 1963, 85, 3533-3539.
7. P. K. Chattaraj and R. G. Parr, in *Structure And Bonding*, ed. K. D. Sen, 1993, vol. 80, ch. 2, pp. 11-25.
8. R. G. Pearson and W. E. Palke, *Journal of Physical Chemistry*, 1992, 96, 3283-3285.
9. P. K. Chattaraj, H. Lee and R. G. Parr, *J Am Chem Soc*, 1991, 113, 1855-1856.
10. R. G. Parr and P. K. Chattaraj, *J Am Chem Soc*, 1991, 113, 1854-1855.
11. P. W. Ayers and R. G. Parr, *J Am Chem Soc*, 2000, 122, 2010-2018.
12. J. L. Gazquez, *J Phys Chem A*, 1997, 101, 4657-4659.
13. P. W. Ayers, *Faraday Discussions*, 2007, 135, 161-190.
14. P. K. Chattaraj, P. W. Ayers and J. Melin, *Phys Chem Chem Phys*, 2007, 9, 3853-3856.
15. P. W. Ayers, R. G. Parr and R. G. Pearson, *J Chem Phys*, 2006, 124, 194107.
16. P. K. Chattaraj and P. W. Ayers, *J Chem Phys*, 2005, 123, 086101.
17. P. W. Ayers, *J Chem Phys*, 2005, 122, 141102.
18. C. Cardenas and P. W. Ayers, *Phys Chem Chem Phys*, 2013, 15, 13959-13968.
19. C. Cardenas and P. Fuentealba, *Concepts and Methods in Modern Theoretical Chemistry: Electronic Structure and Reactivity*, 2013, 65.
20. P. W. Ayers and C. Cardenas, *J Chem Phys*, 2013, 138, 181106.
21. P. Geerlings and F. De Proft, *International Journal of Quantum Chemistry*, 2000, 80, 227-235.
22. G. K. Patra, S. Hati and D. Datta, *Indian Journal of Chemistry Section a-Inorganic Bio-Inorganic Physical Theoretical & Analytical Chemistry*, 1999, 38, 1-3.
23. J. L. Reed, *Inorganic Chemistry*, 2009, 48, 7151-7158.
24. R. G. Pearson, *J.Chem.Educ.*, 1987, 64, 561-567.
25. P. Fuentealba and C. Cardenas, in *Chemical Modelling: Volume 11*, The Royal Society of Chemistry, 2015, vol. 11, pp. 151-174.
26. R. G. Parr and L. J. Bartolotti, *J Am Chem Soc*, 1982, 104, 3801-3803.
27. P. Fuentealba and R. G. Parr, *J Chem Phys*, 1991, 94, 5559-5564.
28. P. W. Ayers, *Journal of Mathematical Chemistry*, 2008, 43, 285-303.
29. C. Cardenas, N. Rabi, P. W. Ayers, C. Morell, P. Jaramillo and P. Fuentealba, *J Phys Chem A*, 2009, 113, 8660-8667.
30. P. Senet, *J Chem Phys*, 1996, 105, 6471-6489.
31. P. Geerlings and F. De Proft, *Phys Chem Chem Phys*, 2008, 10, 3028-3042.
32. J. P. Perdew, R. G. Parr, M. Levy and J. L. Balduz, Jr., *Physical Review Letters*, 1982, 49, 1691-1694.

33. W. T. Yang, Y. K. Zhang and P. W. Ayers, *Physical Review Letters*, 2000, 84, 5172-5175.
34. J. Cioslowski and B. B. Stefanov, *J Chem Phys*, 1993, 99, 5151-5162.
35. A. Cedillo, D. Van Neck and P. Bultinck, *Theoretical Chemistry Accounts*, 2012, 131, 1227.
36. P. W. Ayers, *Theoretical Chemistry Accounts*, 2007, 118, 371-381.
37. R. A. Miranda-Quintana and P. W. Ayers, *The Journal of Chemical Physics*, 2016, 144, 244112.
38. R. A. Miranda-Quintana and P. W. Ayers, *Phys Chem Chem Phys*, 2016, 18, 15070-15080.
39. P. Fuentealba and C. Cardenas, *Journal of Molecular Modeling*, 2013, 19, 2849-2853.
40. F. Heidar-Zadeh, M. Richer, S. Fias, R. A. Miranda-Quintana, M. Chan, M. Franco-Pérez, C. E. González-Espinoza, T. D. Kim, C. Lanssens, A. H. G. Patel, X. D. Yang, E. Vöhringer-Martínez, C. Cárdenas, T. Verstraelen and P. W. Ayers, *Chem Phys Lett*, DOI: <http://dx.doi.org/10.1016/j.cplett.2016.07.039>.
41. S. Noorizadeh and E. Shakerzadeh, *J Phys Chem A*, 2008, 112, 3486-3491.
42. S. Noorizadeh and H. Parsa, *J Phys Chem A*, 2013, 117, 939-946.
43. C. Cardenas, *Chem Phys Lett*, 2011, 513, 127-129.
44. F. Heidar-Zadeh, R. Miranda-Quintana, T. Verstraelen, P. Bultinck, and P. W. Ayers (submitted).
45. P. W. Ayers and M. Levy, *Theoretical Chemistry Accounts*, 2000, 103, 353-360.
46. R. G. Parr and W. T. Yang, *J Am Chem Soc*, 1984, 106, 4049-4050.
47. W. T. Yang, R. G. Parr and R. Pucci, *J Chem Phys*, 1984, 81, 2862-2863.
48. P. W. Ayers, F. De Proft and P. Geerlings, *Physical Review A*, 2007, 75, 012508.
49. C. Morell, A. Grand and A. Toro-Labbe, *J Phys Chem A*, 2005, 109, 205-212.
50. J. F. Capitani, R. F. Nalewajski and R. G. Parr, *J Chem Phys*, 1982, 76, 568-573.
51. F. De Proft, P. W. Ayers, S. Fias and P. Geerlings, *J Chem Phys*, 2006, 125, 214101.
52. P. Geerlings, P. W. Ayers, A. Toro-Labbe, P. K. Chattaraj and F. De Proft, *Accounts of Chemical Research*, 2012, 45, 683-695.
53. P. W. Ayers and R. G. Parr, *J Am Chem Soc*, 2001, 123, 2007-2017.
54. O. A. von Lilienfeld and M. E. Tuckerman, *J Chem Theory Comput*, 2007, 3, 1083-1090.
55. C. Cardenas, P. W. Ayers and A. Cedillo, *J Chem Phys*, 2011, 134, 174103.
56. R. G. Parr and W. Yang, *Density-Functional Theory of Atoms and Molecules*, Oxford UP, New York, 1989.
57. C. Cardenas, P. Ayers, F. De Proft, D. Tozer and P. Geerlings, *Phys Chem Chem Phys*, 2011, 13, 2285-2293.
58. Kramida, A., Ralchenko, Yu., Reader, J. and NIST ASD Team (2015). NIST Atomic Spectra Database (version 5.3), [Online]. Available: <http://physics.nist.gov/asd> National Institute of Standards and Technology, Gaithersburg, MD.
59. D. R. Lide, *CRC Handbook of Chemistry and Physics*, CRC Press, Boca Raton, 1990.

60. N. H. March and R. G. Parr, *Proceedings of the National Academy of Sciences*, 1980, 77, 6285-6288.
61. A. M. Teale, F. De Proft and D. J. Tozer, *J Chem Phys*, 2008, 129, -.
62. F. De Proft, N. Sablon, D. J. Tozer and P. Geerlings, *Faraday Discussions*, 2007, 135, 151-159.
63. D. J. Tozer and F. De Proft, *J Chem Phys*, 2007, 127, 034108.
64. D. J. Tozer and F. De Proft, *J Phys Chem A*, 2005, 109, 8923-8929.
65. N. H. March and R. J. White, *Journal of Physics B: Atomic and Molecular Physics* 1972, 5, 466-475.
66. E. A. Hylleraas, *Zeitschrift für Physik*, 1930, 65, 209-225.
67. E. P. Ivanova and U. I. Safronova, *Journal of Physics B: Atomic and Molecular Physics*, 1975, 8, 1591.
68. U. I. Safronova and A. M. Urnov, *Journal of Physics B: Atomic and Molecular Physics*, 1979, 12, 3171.
69. S. J. Chakravorty and E. R. Davidson, *Journal of Physical Chemistry*, 1996, 100, 6167-6172.
70. J. Linderberg and H. Shull, *Journal of Molecular Spectroscopy*, 1961, 5, 1-16.
71. C. Cardenas, W. Tiznado, P. W. Ayers and P. Fuentealba, *The Journal of Physical Chemistry A*, 2011, 115, 2325-2331.
72. E. Sagvolden and J. P. Perdew, *Physical Review A*, 2008, 77, 012517.
73. A. Gonis, X. G. Zhang, D. M. Nicholson and G. M. Stocks, *Physical Review B*, 2011, 84.
74. R. C. Bochicchio and D. Rial, *J Chem Phys*, 2012, 137.
75. R. C. Bochicchio, R. A. Miranda-Quintana and D. Rial, *The Journal of Chemical Physics*, 2013, 139, 191101.
76. R. A. Miranda-Quintana and R. C. Bochicchio, *Chem Phys Lett*, 2014, 593, 35-39.
77. R. C. Bochicchio, *Theoretical Chemistry Accounts*, 2015, 134, 138.
78. A. Gonis, X. G. Zhang, D. M. Nicholson and G. M. Stocks, *Journal of Physics and Chemistry of Solids*, 2014, 75, 680-687.
79. A. Gonis, X. G. Zhang, D. M. Nicholson and G. M. Stocks, *Molecular Physics*, 2014, 112, 453-461.
80. N. Agmon, *Journal of Chemical Education*, 1988, 65, 42-44.
81. J. H. Harding and N. C. Pyper, *Philosophical Magazine Letters*, 1995, 71, 113-121.
82. H. Hogreve, *Journal of Physics B: Atomic, Molecular and Optical Physics*, 1998, 31.
83. G. Gil, arXiv, 2012, vol. arXiv :1208.6020 [physics.atom-ph].